

DWEX AI-POWERED GROUNDWATER MODELLING WORKFLOW

Target: Managing Director Scope: AI-Assisted MODFLOW 6 Groundwater Modelling Framework
 Status: R&D Proposal

PAGE 1 — PHASE 1: RAW DATA TO VALIDATED MODEL SETUP

Step 1 — Raw DWEX Project Data

The project team provides the complete available groundwater/project dataset as the immutable technical baseline.

- Soil and geotechnical reports; borehole logs and lithology; hydraulic/permeability and aquifer-test data; groundwater monitoring records; well and pumping data; rainfall/recharge/ET information where relevant; DEM/topography; GIS data and site coordinates; rivers/drains/lakes or coastal boundary data where relevant; site CAD drawings and existing groundwater studies; project objectives and required scenarios.

Principle: No measured/project parameter is silently fabricated by the AI.

Step 2 — AI Data Inventory & Extraction

The AI agent scans the supplied files and identifies model-relevant information.

- Coordinates and CRS; ground/surface elevations; lithology and geological contacts; aquifer/aquitard information; hydraulic conductivity and storage parameters; piezometric/groundwater heads; well locations, screens and pumping rates; recharge and boundary information; relevant engineering constraints.

Every extracted value maintains a source reference such as document, page, table, figure or dataset location.

Step 3 — Structured Engineering Dataset

Extracted information is converted into a unified machine-readable dataset.

- Value; unit; source/reference; coordinate/CRS where applicable; confidence; status; whether measured, derived or assumed.

Integrity rule: Unknown or unverified parameters remain NULL and are explicitly flagged. The AI must not present invented values as project data.

Step 4 — GIS & Spatial Validation

The AI prepares and validates the spatial foundation before numerical modelling.

- CRS identification and consistency; coordinate validity; project boundary; DEM coverage; borehole/well locations; river/drain/coastal geometry; spatial overlap and alignment; mapping of important project features to real-world coordinates.

The original project coordinates are preserved; transformations are recorded and reversible.

Step 5 — Conceptual Groundwater Model

Using the validated data, the AI prepares the conceptual model for engineering review.

- Geological/aquifer layers; aquitards and bedrock; recharge and discharge mechanisms; groundwater levels; groundwater flow system; wells and pumping stresses; surface-water or coastal interactions; proposed model boundaries; steady-state/transient modelling requirements.

Step 6 — Project Engineer Validation & Approval Gate

The Project Engineer reviews the extracted values, geological interpretation, derived parameters, boundary conditions, initial heads, model extent/grid concept, missing or uncertain data, and conceptual assumptions.

- Engineer corrects or supplies required information.

Mandatory approval gate: MODFLOW numerical input compilation cannot proceed until the engineer approves the engineering dataset and conceptual model.

PAGE 2 — PHASE 2: MODFLOW 6, QA/QC, CALIBRATION & VISUALIZATION

Step 7 — MODFLOW 6 Numerical Model Generation

The AI uses Python/FloPy to convert the engineer-approved dataset into native MODFLOW 6 input files.

- Simulation/name files; discretization/grid; initial conditions; hydraulic properties; recharge; wells/pumping; constant/general-head or other applicable boundaries; solver settings; output-control settings.

Important: MODFLOW 6 is the numerical groundwater engine. The AI does not replace the groundwater physics calculations.

Step 8 — MODFLOW 6 Simulation

The AI agent executes the local `mf6.exe` engine using the generated input files.

- MODFLOW 6 performs the numerical groundwater-flow calculations and produces hydraulic heads, cell-by-cell flows, water-budget information, and simulation/convergence logs.

The simulation is treated as an iterative engineering workflow, not as an automatic final answer.

Step 9 — Automated QA/QC & Iteration

The AI reads the MODFLOW output and performs automated checks.

- Successful execution; solver/convergence status; water-budget consistency; unrealistic/anomalous heads; dry/problematic cells; boundary-condition issues; spatial consistency; numerical instability.

If a problem is detected, the AI identifies the likely cause, proposes/implements a permitted correction, regenerates the input files, reruns `mf6.exe`, and repeats the QA/QC cycle. Critical engineering decisions remain subject to Project Engineer approval.

Step 10 — Calibration

Where sufficient observed groundwater data exists, the AI compares observed vs simulated groundwater levels, spatial residuals, temporal hydrographs, and calibration statistics.

- Controlled adjustment of approved calibration parameters; iterative MODFLOW 6 reruns; calibration statistics and residual assessment.

Data limitation rule: If calibration data is insufficient, the system explicitly reports that calibration could not be completed rather than manufacturing observations.

Step 11 — Independent Validation

The calibrated model is tested against independent observations or a separate time period where available.

- Head agreement; residual behaviour; hydrographs; water balance; physical plausibility; performance against predefined engineering acceptance criteria.

The Project Engineer provides final technical acceptance.

Step 12 — Engineering Visualization

Validated MODFLOW results are converted into spatial and analytical outputs using Python/GIS/visualization tools.

- **Maps:** hydraulic-head, groundwater-elevation, drawdown, recharge/discharge, wells and boundary overlays.
- **Contours:** head, drawdown and groundwater-elevation contours.
- **Graphs:** observed vs simulated heads, hydrographs, residual plots, calibration statistics, water-budget plots, pumping vs drawdown.
- **Cross-sections:** geological layers, aquifer/aquitard structure, groundwater levels, head distribution, flow behaviour.
- **3D:** ground surface/terrain, geological layers, groundwater surface, wells, model domain and drawdown response.

All spatial visualizations retain the project's validated coordinate reference system and spatial relationships.

Step 13 — Scenario Modelling

The validated base model becomes the controlled starting point for engineering scenarios.

- Existing pumping; increased pumping; construction dewatering; multiple-well pumping; future recharge conditions; other project-specific conditions.

Each scenario is run through MODFLOW 6 and receives its own outputs, maps, contours, graphs and comparisons against the base case.

FINAL MODEL & DELIVERABLES

Step 14 — Final Model & Deliverables

The final project package contains the complete traceable modelling workflow:

```
FINAL_GROUNDWATER_MODEL/  
■  
■ ■ ■ ■ 01_Raw_Data/  
■ ■ ■ ■ 02_Extracted_Data/  
■ ■ ■ ■ 03_GIS/  
■ ■ ■ ■ 04_Conceptual_Model/  
■ ■ ■ ■ 05_MODFLOW/  
■ ■ ■ ■ Input_Files/  
■ ■ ■ ■ Output_Files/  
■ ■ ■ ■ 06_Calibration/  
■ ■ ■ ■ 07_Validation/  
■ ■ ■ ■ 08_Scenarios/  
■ ■ ■ ■ 09_Visualizations/  
■ ■ ■ ■ Maps/  
■ ■ ■ ■ Contours/  
■ ■ ■ ■ Graphs/  
■ ■ ■ ■ Cross_Sections/  
■ ■ ■ ■ 3D/  
■ ■ ■ ■ 10_Final_Report/
```

- Native MODFLOW 6 input files
- MODFLOW 6 output files
- Validated engineering dataset
- Calibration and validation results
- Maps and contour plots
- Graphs and hydrographs
- Cross-sections and 3D visualizations
- Scenario results
- Assumptions and data-quality record
- Final technical report

CORE SYSTEM ARCHITECTURE

```
PROJECT DATA  
↓  
AI AGENT + IDE  
↓  
DATA EXTRACTION & STRUCTURING  
↓  
GIS / CRS VALIDATION  
↓  
CONCEPTUAL MODEL  
↓  
ENGINEER APPROVAL  
↓  
Python + FloPy  
↓  
MODFLOW 6 INPUT FILES  
↓  
mf6.exe  
↓  
MODFLOW OUTPUT FILES  
↓  
AI QA/QC  
↓  
ITERATION  
↓  
CALIBRATION  
↓  
VALIDATION  
↓  
MAPS + CONTOURS + GRAPHS  
↓  
CROSS-SECTIONS + 3D  
↓  
SCENARIO MODELLING  
↓  
FINAL VALIDATED MODEL + REPORT
```

GOVERNANCE PRINCIPLE

- **AI:** Data extraction, structuring, coding, automation, QA/QC assistance, visualization and reporting.
- **Python/FloPy:** Programmatic bridge between the engineering dataset and MODFLOW 6.
- **MODFLOW 6 (mf6.exe):** Numerical groundwater-flow simulation engine.
- **GIS/Visualization tools:** Spatial processing and presentation of model results using validated project coordinates.
- **Project Engineer:** Final authority for data acceptance, conceptual assumptions, calibration/validation and engineering conclusions.

AI accelerates and standardizes the modelling workflow; it does not replace engineering accountability or numerical groundwater physics.